



# APPLICATION FLOW DIAGRAM

## FOR FACE MASK DETECTION

### == USING MOBILENETV2 ==

### IN DEEP LEARNING



1



#### WEB CAMERA

Initialize and start the video stream.

2



#### CAPTURE VIDEO FRAME

Capture frame from the live video.

3



#### FACE DETECTION

Detect faces in the frame using OpenCV DNN SSD.

4



#### EXTRACT FACE REGION (ROI)

Extract the detected face region.

5



#### IMAGE RESIZE

Resize the face image to 224 × 224.

6



#### PREPROCESSING

Preprocess the image using MobileNetV2 preprocess.

7



#### LOAD TRAINED MODEL

Load the trained model (mask\_detector.model).

8



#### MASK PREDICTION

Predict whether the person is wearing a mask or not.

9



#### DECISION MAKING

Make decision based on prediction result.

#### MASK DETECTED



GREEN BOUNDING BOX DISPLAY

“ALLOWED” MESSAGE

#### NO MASK DETECTED



RED BOUNDING BOX DISPLAY

“NOT ALLOWED” MESSAGE

#### DISPLAY OUTPUT ON SCREEN

Mask: You are allowed



No Mask: You are not allowed



#### KEY BENEFITS



Real-time Detection



High Accuracy



Fast Performance



Contactless Monitoring



Suitable for Various Environments



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